



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science and Engineering

Natural Language Processing Laboratory

Course Code: CSE 4122

Project Proposal

Implicit Crisis Language Detection Using Transformer-Based NLP

**Distinguishing Explicit, Implicit and Hard-Negative
Crisis Language with a Fine-Tuned BERT Classifier**

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1. Introduction

Computational work on social-media distress is largely organised around *explicit* disclosure. Models trained on such data inherit a lexical shortcut: they learn that terms like “suicide” or “kill myself” predict the positive class, and this shortcut fails in two directions. Distress is frequently expressed *implicitly* — through hopelessness, perceived burdensomeness or farewell-like phrasing — with no crisis vocabulary at all; and crisis vocabulary appears abundantly in text expressing no personal crisis, such as news commentary, third-party concern and hyperbole.

Compare “I want to kill myself” (explicit) with “Everyone would be better off without me” (implicit: no crisis term occurs, and meaning depends on composition in context). A bag-of-words representation can only encode the first; contextual transformer representations should, in principle, encode the second.

Research question. Can transformer-based NLP models identify crisis-related language expressed implicitly, when explicit suicide-related terminology is absent?

Scope. A text-classification research prototype, not a clinical diagnostic system; outputs classify language, not people.

2. Research Gap

The gap addressed is an evaluation and annotation gap, not a claim of novel subject matter.

- **Labelling provenance.** Widely used corpora derive labels from subreddit membership rather than post content, introducing label noise invisible in aggregate accuracy.
- **Explicit dominance.** Where positives come from disclosure-norm communities, explicit phrasing is over-represented, so a model can score highly while learning little beyond a keyword list.
- **Absence of hard negatives.** Negatives are usually topically unrelated, so the boundary is never tested against crisis vocabulary lacking personal crisis.
- **No four-way organisation.** No surveyed dataset separates Explicit / Implicit / Hard Negative / Non-Crisis, so these slices cannot be reported independently.

Our own corpora confirm this. In the IRF splits (3,522 posts), 73.2% of risk-labelled posts contain none of the markers *suicid**, *kill myself*, *end my life*; in SDCNL (1,895 posts), 62.8% of positives contain no such marker while 18.4% of negatives do. Implicit and hard-negative material is already latent here, but is neither separately labelled nor evaluable.

3. Proposed Contribution

- A manually annotated evaluation layer over existing public corpora distinguishing **Explicit Crisis**, **Implicit Crisis**, **Hard Negative** and **Non-Crisis** — the primary contribution.
- A controlled comparison of TF-IDF + Logistic Regression against a fine-tuned BERT classifier on identical data and an identical unseen test set.
- Disaggregated evaluation reporting explicit, implicit and hard-negative cases separately rather than as one pooled score.
- A qualitative error analysis of where contextual representation helps and which hard negatives induce false positives.

4. Dataset Sources

Only the manually annotated gold dataset provides labels for our four target categories; the corpora below supply candidate text and conceptual grounding.

Dataset	Role	Size	Labels
Komati (Reddit)	Candidate text; bulk data	~232k	Binary suicide / non-suicide, subreddit-derived
IRF (ACL '23)	Implicit-risk candidates	3,522	Thwarted Belongingness, Perceived Burdensomeness
SDCNL (ICANN '21)	Hard-negative candidates	1,895	Binary is_suicide (980 / 915)
CAMS (LREC '22)	Implicit distress via cause	5,052	Six causal categories + spans
GoEmotions (ACL '20)	Non-crisis bulk text	58,009	27 emotion labels + Neutral

Note on IRF. Its constructs frequently surface as implicit crisis language, so it calibrates our guidelines and locates candidates; it is not itself treated as a crisis dataset.

5. Proposed Methodology

1. **Collect candidates.** Sample the corpora above, over-sampling posts with risk or causal annotations but no explicit vocabulary, and posts with crisis vocabulary but a negative label.
2. **Human annotation.** Each post receives exactly one of the four labels under written guidelines; a portion is double-annotated so inter-annotator agreement can be reported.
3. **Gold dataset.** Annotated items form the gold set; a target of ~2,000 posts is indicative, not finalised.
4. **Split.** Partition into train/validation and a held-out test set, untouched during tuning and stratified so all three slices are represented.
5. **Models.** A baseline of TF-IDF features (word and character n-grams) with Logistic Regression, and a pre-trained BERT fine-tuned on the same data — both tuned on the same validation split.
6. **Evaluation.** Both models on the identical test set: accuracy, precision, recall, F1 and confusion matrices overall, and — as the primary result — computed separately over explicit, implicit and hard-negative cases.
7. **Error analysis.** Examine items where TF-IDF succeeds and BERT fails, the reverse, where both fail, and hard negatives causing false positives.

6. Ethical Considerations

All data is drawn from existing publicly released research corpora with author identifiers removed; no user is identified, profiled or contacted, no clinical diagnosis is performed, and the system is never deployed on live user data.

7. Limitations

Subreddit-derived labels are noisy candidate signals rather than ground truth; the gold dataset is small relative to corpora typically used for fine-tuning; the implicit / non-crisis boundary is subjective, so annotator disagreement is expected and will be reported rather than concealed; and findings are limited to English social-media text, where classification performance is not clinical validity.